Subject Description Form

Subject Code	AAE5103
Subject Title	Artificial Intelligence in Aviation Industry
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject will provide students with 1. the main concepts, ideas and techniques of advanced artificial intelligence (AI) in the aviation industry; 2. the essential principles, research methodology, data interpretation and data analysis with case examples in airline and airport operations; and 3. outlook of artificial intelligence development and its important in future air traffic and unmanned aircraft system traffic management.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. design and develop AI algorithms or adopt AI tools in solving engineering problems in airline and airport operations; b. illustrate and analysis the knowledge and data pattern generated by the AI-engine; c. master and understand the complex causal relationship and inferences of AI; and d. apply AI techniques for solving aviation engineering problems.
Subject Synopsis/ Indicative Syllabus	**Fundamental of Artificial Intelligence (AI)**: Basic concepts and trends of AI; AI algorithm design; deep learning; large language models. **Data Analytics:** Data processing; data mining; data visualization; principle component analysis; data-driven optimization. **Supervised learning:** Least squares and nearest neighbours; statistical decision theory; Linear methods for regression; Linear discriminant analysis; Classifications; Logistic regression; Support-vector machine. **Unsupervised learning:** Clustering; Association dimensionality reduction; K-means clustering; KNN. **Model inference and averaging:** Bootstrap and maximum likelihood methods; Bayesian method. **Reinforcement learning:** Basic concepts of reinforcement learning; Markov Decision Processes (MDP); Q-learning. **Applications in Aviation:** Air traffic demand forecasting; Flight delay prediction; Operations management and dynamic pricing; managerial implications and actionable insights with aviation case studies analysis.

Teaching/Learning Methodology

Teaching is conducted through lectures and case studies. The basic knowledge, research methodology and theoretical models will be introduced. The understanding of how to address and formulate problems using AI models and soft computing techniques is emphasised. Research methodology, data analytics skills, and algorithm design skills are taught in class as well as the related real-life scenarios using data to enhance their research abilities.

Teaching/Learning Methodology	Outcomes			
	a	b	c	d
Lecture	√	√	√	√
Case Study	√	√	√	√

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
		a	b	c	d
1. Assignment	20%	√	√		
2. Project report	20%		√	√	√
3. Project presentation	10%		√	√	√
4. Final examination	50%	√	√	√	√
Total	100%				

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Overall Assessment:

$$0.5 \times \text{Continuous Assessment} + 0.5 \times \text{Final Examination}$$

The continuous assessment (50%) is aimed at enhancing the students' comprehension and assimilation of various topics of the syllabus via assignment and group project. The final examination (50%) will also be considered to assess the students' learning outcome, with a focus on their understanding of basic concepts, AI models, and application methods.

Student Study Effort Expected

Class contact:	
▪ Lecture/Case Study	39 Hrs.
Other student study effort:	
▪ Literature review/case study/reading	36 Hrs.

	▪ Self-study/preparation	36 Hrs.
	Total student study effort	111 Hrs.
Reading List and References	1. Barber, D. (2012). Bayesian reasoning and machine learning. Cambridge University Press. 2. Boyd, S., Boyd, S. P., & Vandenberghe, L. (2004). Convex optimization. Cambridge university press. 3. Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2009). Introduction to algorithms. MIT press. 4. De Neufville, R., & Odoni, A. (2003). Airport systems. planning, design and management. New York: McGraw-Hill. 5. EASA (2020). EASA Artificial Intelligence Roadmap 1.0 published: A human-centric approach to AI in aviation. EASA. 6. Eurocontrol. (2020). FLY AI report – demystifying and accelerating AI in aviation/ATM. Eurocontrol. 7. Guido, S., & Müller, A. (2016). Introduction to machine learning with python (Vol. 282). O'Reilly Media. 8. Marsland, S. (2015). Machine learning: an algorithmic perspective. CRC press. 9. Richert, W. (2013). Building machine learning systems with Python. Packt Publishing Ltd.	

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